

MOUNTIAN EMPIRE, VA, USA

WMO: 724056

Lat: 36.895N

Lon: 81.350W

Elev: 780

StdP: 92.30

Time Zone: -5.00 (NAE)

Period: 95-19

WBAN: 63805

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.9	-10.4	-19.5	0.7	-8.3	-17.2	0.9	-7.9	14.0	-0.6	12.7	-1.6	5.2	270	0.454

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.6	31.8	21.5	30.1	21.2	28.8	20.7	23.3	28.0	22.6	27.3	21.9	26.8	2.6	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.1	18.5	25.2	21.3	17.5	24.6	20.5	16.7	24.0	73.6	27.9	70.8	27.6	68.1	27.1	27.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.5	9.9	8.5	-16.2	32.6	3.8	2.0	-18.9	34.1	-21.2	35.3	-23.3	36.4	-26.1	37.9		
(5)			-16.8	24.6	3.6	1.1	-19.4	25.4	-21.5	26.0	-23.5	26.6	-26.1	27.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	12.4	0.9	2.8	7.2	12.3	17.0	21.0	22.8	22.2	18.8	12.9	6.7	3.3		
	DBStd	8.97	6.56	5.90	5.77	4.92	3.87	2.73	2.42	2.52	3.50	4.85	5.00	5.40		
	HDD10.0	1024	290	211	123	33	3	0	0	0	1	27	121	216		
	HDD18.3	2623	541	436	346	187	71	10	2	3	37	176	348	466		
	CDD10.0	1889	7	8	36	101	219	329	398	379	263	117	23	8		
	CDD18.3	446	0	0	1	4	29	89	142	123	49	8	0	0		
	CDH23.3	3520	0	1	10	66	266	666	1060	953	426	69	3	0		
	CDH26.7	956	0	0	1	4	40	169	322	298	112	9	1	0		
(14) Wind	WSAvg	3.1	4.6	4.3	3.9	3.6	2.7	2.2	2.2	1.7	1.9	2.7	3.3	3.9		
(15) Precipitation	PrecAvg	1089	90	68	107	82	102	110	115	101	87	69	74	86		
	PrecMax	1424	205	136	201	148	199	213	197	184	309	231	157	187		
	PrecMin	847	32	18	31	15	33	38	46	27	8	1	15	38		
	PrecStd	157	46	31	41	33	49	49	36	42	61	55	40	34		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.0	21.1	24.9	27.3	29.7	32.3	32.9	32.9	32.3	27.8	22.6	19.3		
		MCWB	14.2	13.3	14.9	16.5	20.1	21.3	22.4	22.0	19.8	18.7	14.6	14.1		
	2%	DB	16.2	17.3	22.0	25.3	27.7	30.1	31.3	31.4	29.5	25.3	19.1	16.6		
		MCWB	11.6	10.6	13.9	15.8	18.6	20.9	21.8	21.6	20.1	17.4	13.0	12.5		
	5%	DB	13.0	14.1	18.8	23.2	26.3	28.6	29.9	29.9	27.7	22.9	17.6	14.0		
		MCWB	10.0	9.4	11.9	14.7	18.0	20.4	21.8	21.1	19.1	15.8	12.1	10.7		
	10%	DB	11.0	12.1	16.8	21.3	24.7	27.3	28.4	28.0	26.1	21.2	15.7	12.2		
		MCWB	8.2	8.2	10.9	13.9	17.5	19.7	21.1	20.5	18.8	15.7	11.3	9.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.8	15.1	16.1	18.6	21.6	23.2	24.7	23.9	22.7	21.4	16.3	15.6		
		MCDB	17.9	18.7	22.3	24.0	26.4	27.7	29.5	29.3	27.4	24.5	20.3	18.0		
	2%	WB	12.8	13.0	14.9	17.2	20.4	22.3	23.7	23.0	21.4	19.4	15.0	13.5		
		MCDB	15.5	16.1	19.9	23.1	25.3	27.3	28.2	28.2	26.0	23.1	18.1	15.7		
	5%	WB	10.7	10.6	13.5	16.0	19.4	21.5	22.9	22.3	20.5	17.7	13.6	11.7		
		MCDB	12.9	13.2	17.9	21.4	24.4	26.7	27.3	27.5	25.4	21.6	16.6	13.5		
	10%	WB	8.2	8.5	12.0	14.9	18.5	20.7	22.1	21.6	19.6	16.3	11.6	9.5		
		MCDB	10.3	11.1	15.7	19.8	23.2	25.6	26.7	26.7	24.6	19.9	14.8	11.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.4	9.8	11.7	12.5	12.1	11.3	10.6	11.3	11.9	12.2	11.6	9.4		
		MCDBR	12.3	14.2	15.3	15.6	14.4	13.6	12.5	13.8	14.8	15.2	15.2	12.9		
	5% WB	MCWBR	9.0	9.4	8.8	8.1	7.3	6.2	5.6	6.1	6.8	8.2	9.4	9.0		
		MCWBR	10.9	11.6	12.9	12.6	11.7	11.1	10.4	11.2	12.0	11.1	11.5	10.5		
(40) Clear-Sky Solar Irradiance	taub		0.290	0.297	0.321	0.364	0.404	0.433	0.464	0.469	0.367	0.334	0.301	0.283		
	taud		2.521	2.488	2.468	2.363	2.250	2.197	2.142	2.103	2.386	2.481	2.544	2.564		
	Ebn at Noon		911	945	947	917	880	851	820	808	885	889	890	898		
	Edh at Noon		78	91	103	121	137	145	152	155	110	91	76	71		
(44) All-Sky Solar Radiation	RadAvg		2.22	2.85	3.97	5.04	5.55	6.02	5.61	5.22	4.45	3.49	2.61	1.93		
	RadStd		0.19	0.35	0.26	0.40	0.43	0.41	0.37	0.26	0.41	0.43	0.26	0.18		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (42 neighbors)		+0.45	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+138	+82	

Nomenclature: See separate page